

ANANYA SRIDHAR

AI/ML Engineer — Applied AI, Speech AI, Computer Vision & Robotics

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(<https://linkedin.com/in/>)

Languages: English, Hindi, Tamil

PROFESSIONAL SUMMARY

Full-Stack AI/ML Engineer with a proven track record of architecting and deploying end-to-end production systems across **Speech AI, Agentic Systems (LangGraph, RAG), Computer Vision, and Autonomous Robotics**. Experienced in taking complex machine learning prototypes—from physics-backed MuJoCo simulations and vision transformer optimal transport to domain-fine-tuned legal LLMs—into scalable, low-latency, client-facing applications. Skilled in high-performance backends (FastAPI, WebSockets, PostgreSQL), modern web frameworks (React, Vite, Next.js, Three.js), and edge/CPU optimization.

TECHNICAL SKILLS

- Languages:** Python (Proficient), C++, C, SQL, JavaScript (ES6+), TypeScript
- AI / Deep Learning:** PyTorch, Hugging Face (Transformers, PEFT, Accelerate), LangGraph, LangChain, RAG, ChromaDB, Reinforcement Learning (DRL, PPO, Q-Learning), MCTS, A* Pathfinding, Scikit-Learn
- Speech & Audio AI:** Faster-Whisper, OpenAI Whisper, Vosk, Coqui TTS, Kokoro ONNX, Librosa, Praat-Parselmouth, Epitran (G2P), Acoustic Feature Extraction, Phoneme Verification
- Computer Vision:** OpenCV, MediaPipe Tasks Vision, DINOv2 (Vision Transformers), Optimal Transport (POT), Background Subtraction, Multi-Object Centroid Tracking, One-Shot Face Recognition, LIME / SHAP
- Robotics & Simulation:** MuJoCo 3D Physics Simulation, Gymnasium, Sensor Fusion (Extended Information Filter - EIF), Three.js, React Three Fiber, Forward & Inverse Kinematics, Lidar Motion Planning
- Backend & Cloud:** FastAPI, Uvicorn, PostgreSQL, SQLite, Supabase, Azure Cloud, Docker, WebSockets, RESTful APIs, Alembic, Redis
- Frontend & Tooling:** React.js, Vite, Next.js (App Router), Tailwind CSS, HTML5 Canvas, PWA, Git, Linux, Bash

PROFESSIONAL EXPERIENCE

AI/ML ENGINEER, FULL-STACK | MANAS LEARNING

Mar 2026 – Present | Remote | Mumbai, India

- End-to-End Platform Architecture:** Architected and deployed a full-stack pediatric speech therapy platform (FastAPI, PostgreSQL, React + Vite) serving multi-language clinical assessments across 8 Indian languages.
- LangGraph Speech Assessment Pipeline:** Engineered a real-time pronunciation scoring engine chaining Faster-Whisper, Vosk, and Kokoro/Coqui TTS with phoneme-level acoustic verification (Librosa, Epitran G2P) for instant articulatory feedback.
- Reinforcement Learning Adaptive Engine:** Designed and implemented a DRL/Q-learning adaptive difficulty controller that dynamically calibrates gameplay difficulty per child based on rolling pronunciation accuracy and response latency.
- Visual Biofeedback & Safety:** Integrated MediaPipe facial landmark detection for real-time oral-motor exercise tracking; implemented COPPA-compliant parental consent, secure session tokens, role-based dashboards (therapist vs. parent), and automated clinical progress reports.

AI/ML ENGINEER | NIVESH AI

Jun 2026 – Present | Mumbai, India

- Enterprise Speech AI Infrastructure:** Built end-to-end audio ingestion and intelligence pipelines deployed on Azure for a wealth management platform serving 10,000+ mutual fund distributors (MFDs).
- Domain-Specific STT Optimization:** Integrated domain-adapted speech-to-text models (Sarvam AI, Whisper) with noise-reduction preprocessing (noisereduce), achieving high transcription fidelity on multilingual Indian English / Hindi advisor conversations.
- RAG-Driven Meeting Summarization:** Developed a deterministic, zero-hallucination RAG summarization layer parsing complex advisory call notes and portfolio reviews into structured CRM action items using `pdfplumber`, `python-docx`, and vector embeddings.

FEATURED PROJECTS

NIRVANA (FIREBOT) — AUTONOMOUS FIREFIGHTING ROBOT & MUJOCO SIMULATION

GitHub Repository · Live Documentation & Console

Python, MuJoCo Physics, PyTorch, Gymnasium, EIF Sensor Fusion, FastAPI, WebSockets

- Built an autonomous firefighting robot operating in a physics-backed MuJoCo 3D simulator across procedural 8-room environments with volumetric particle fire dynamics.
- Implemented **MM-FusionRL** (multimodal cross-attention deep RL) coupled with **Extended Information Filter (EIF)** sensor fusion for thermal source localization, lidar SLAM exploration, and obstacle avoidance.
- Developed a high-performance web console streaming real-time robot telemetry, lidar sweeps, and automated fire-event database logs via FastAPI and WebSockets.

TRIPWIRE — REAL-TIME CPU-ONLY COMPUTER VISION VEHICLE COUNTER

GitHub Repository · Live Demo

Python, OpenCV, FastAPI, React, Vite, WebSockets, Docker

- Engineered an edge-ready, CPU-only vehicle detection and tracking pipeline from live CCTV and RTSP feeds with sub-50ms latency.

- Implemented Gaussian background subtraction, morphological noise removal, multi-object centroid tracking, and directional virtual tripwire crossing algorithms.
- Built a low-latency web monitoring dashboard in React and Vite streaming processed video frames via MJPEG and crossing counts via WebSockets.

GECO-ENHANCED — CONFIDENCE-AWARE SEMANTIC KEYPOINT CORRESPONDENCE

[GitHub Repository](#)

PyTorch, DINOv2 (ViT), Python Optimal Transport (POT), FastAPI, React, Docker

- Formulated an unsupervised semantic correspondence framework matching keypoints across different instances of identical object classes under large viewpoint and appearance shifts.
- Extracted dense features using **DINOv2** vision transformers and solved matching via confidence-weighted **Optimal Transport (POT)** with Sinkhorn regularization.
- Shipped an interactive dual-image visualizer allowing real-time pixel-level probe matching and PCA feature map inspection.

AEGIS — MULTI-AGENT DISASTER RESPONSE SIMULATION

[GitHub Repository](#)

*Python, NetworkX, NumPy, A Pathfinding, MCTS, Reinforcement Learning, FastAPI, React**

- Developed a multi-agent simulation coordinating autonomous drone swarms, medical rescue teams, and transport vehicles under extreme time, weather, and battery constraints.
- Engineered hazard-aware *A routing **and** Monte Carlo Tree Search (MCTS)** for dynamic mission assignment, casualty triage, and real-time obstacle replanning.
- Built a WebSocket-driven command console visualizing agent trajectories, casualty status, and disaster map topology.

NYAYAI & AI BAIL RECKONER — INDIAN LEGAL-REASONING LLM & RAG SYSTEM

[GitHub Repository: NyayAI](#) · [GitHub Repository: Bail Project](#)

PyTorch, Hugging Face Transformers, PEFT/LoRA, ChromaDB, FastAPI, React

- Created a specialized legal AI assistant predicting bail hearing outcomes and citing judicial precedents across the Indian Penal Code (IPC) and Bharatiya Nyaya Sanhita (BNS).
- Implemented a dense retrieval RAG pipeline using ChromaDB to ground predictions strictly in verified High Court and Supreme Court rulings, eliminating hallucinations.
- Trained attention-based neural networks on structured legal corpora to extract procedural arguments and judicial rationale.

3-DOF ROBOTIC MANIPULATOR SIMULATOR — KINEMATICS & OPTIMIZATION

[GitHub Repository](#) · [Live Demo](#)

Python, FastAPI, NumPy, SciPy, Three.js, React, Vite, Recharts

- Developed an interactive 3D simulation for 3-DOF planar and spatial robotic arms (RRR and PPP configurations).
- Implemented forward and inverse kinematics solvers using damped least-squares and SciPy numerical optimization for precision end-effector trajectory tracking.
- Visualized reachable workspace boundaries, manipulability ellipsoids, and joint torque limits in real-time Three.js.

REAL-TIME MISINFORMATION PROFILER — EXPLAINABLE NLP FRAMEWORK

[GitHub Repository](#)

Hugging Face Transformers, DistilBERT, BART, LIME, SHAP, Next.js, Socket.io, FastAPI

- Built an explainable AI pipeline detecting viral misinformation on social media by streaming live Reddit discourse via Socket.io.
- Fine-tuned DistilBERT and BART models for topic-sensitive stance detection, coupled with **LIME and SHAP** attribution algorithms to deliver word-level explainability heatmaps for end users.

BESHMAI GRAM — 3D MEMORY PALACE & CLIENT-SIDE NLP JOURNAL

[GitHub Repository](#) · [Live Demo](#)

Next.js (App Router), React Three Fiber, Three.js, Gemini API, Supabase, Xenova Transformers

- Designed an interactive 3D spatial memory palace and reflective journaling application integrating Google Gemini for multimodal cognitive synthesis.
- Implemented on-device, zero-latency sentiment classification using **Xenova Transformers / ONNX** in WebAssembly directly inside the browser, preserving complete user privacy.

BREATHQUEST & IN THEIR WORLD — HEALTHCARE & NEURODIVERSITY PLATFORMS

[BreathQuest Live Demo](#) · [In Their World Live Demo](#)

React, Vite, Web Audio API, HTML5 Canvas, FastAPI, PostgreSQL

- **BreathQuest:** Built 6 gamified HTML5 canvas breath-control games for respiratory therapy, analyzing microphone audio dynamics in real time.
- **In Their World:** Created four interactive sensory simulations (Autism, ADHD, Dyslexia, Speech differences) calibrated with empirical Indian demographic and clinical prevalence statistics.

EDUCATION

Bachelor of Technology (B.Tech)

SASTRA University, India

Expected Graduation: 2027

CERTIFICATIONS & RECOGNITIONS

- **SWAYAM – IIT Madras:** Comprehensive Electronics and Embedded Systems (*In-Person Practical Program*)

- **IIoT Certification:** Ministry of Heavy Industries (MHI) Program (*Project-Based Evaluation*)